WEEK 4 ASSESSMENT

Data + ML Fundamentals Review

Quiz • Preprocessing Task • Concept Viva

Date	28-Oct-2026	**Duration**	4 hours
Difficulty	Intermediate	**Total marks**	100

Assessment Instructions

- Complete independently. Use notes only if the trainer permits them.
- Write short explanations where requested and do not copy code without understanding it.
- You may be asked to explain any line of your notebook during the viva.
- Submit quiz answers and your preprocessing notebook.
- Suggested timing: Quiz 45 minutes; practical task 2 hours; debugging, reflection and viva 1 hour 15 minutes.

Part A — Multiple-Choice Quiz (20 marks)

Choose one best answer for each question. Each question carries 1 mark.

Which statement best describes Machine Learning?

- A. A system learns patterns from examples
- B. A program only follows fixed rules
- C. A database stores files
- D. A network connects devices

Predicting whether a transaction is fraudulent or legitimate is usually:

- A. Regression
- B. Classification
- C. Clustering
- D. Data compression

Predicting a numeric house selling price is usually:

- A. Classification
- B. Clustering
- C. Regression
- D. One-hot encoding

Grouping customers by behaviour without existing group labels is:

- A. Supervised regression
- B. Supervised classification
- C. Unsupervised clustering
- D. Scaling

In a dataset used to predict churn, the column churn is the:

- A. Scaler
- B. Label / target
- C. Test set

- D. Imputer

Which is a possible feature for predicting house price?

- A. The price to be predicted
- B. Number of bedrooms
- C. The model's final score
- D. The prediction itself

Why keep a test set separate?

- A. To enlarge the dataset
- B. To estimate performance on unseen data
- C. To fill missing values
- D. To encode categories

A missing-value imputer is used for:

- A. Handling missing entries
- B. Splitting rows
- C. Choosing the target automatically
- D. Measuring accuracy

A common strategy for a missing numeric feature is:

- A. Replace values with the column name
- B. Use the median learned from training data
- C. Copy values from the target
- D. Convert numbers to city names

StandardScaler generally:

- A. Converts categories to text
- B. Standardises numeric values using learned statistics
- C. Deletes test rows
- D. Predicts labels

A common method to encode nominal data such as city is:

- A. One-hot encoding
- B. Median imputation
- C. Train/test split
- D. Accuracy scoring

OneHotEncoder(handle_unknown='ignore') helps when:

- A. All values are missing
- B. A category appears during transform that was not seen during fitting
- C. There are no features
- D. A model overfits

Data leakage occurs when:

- A. A notebook is not saved
- B. Held-out or unavailable information improperly influences training/evaluation
- C. A dataset has many columns
- D. Accuracy is low

Imputers and scalers should generally be fitted:

- A. On all rows before splitting
- B. On the test set only
- C. On the training set only
- D. On unrelated data

In scikit-learn, fit usually means:

- A. Learn parameters or rules from data
- B. Display a chart
- C. Remove the target automatically
- D. Save a notebook

In scikit-learn, transform usually means:

- A. Apply a learned transformation
- B. Learn from test labels
- C. Create business requirements
- D. Split rows randomly

A benefit of a scikit-learn Pipeline is that it:

- A. Guarantees 100% accuracy
- B. Combines steps and applies them consistently
- C. Removes evaluation needs
- D. Makes all missing values harmless

Which problem may be better solved with a simple rule?

- A. Flag an invoice number that duplicates one for the same vendor
- B. Predict house price from many historical examples
- C. Discover customer groups from behaviour
- D. Forecast sales from trends

Training and evaluating on the same rows can mislead because:

- A. Labels are always removed
- B. It may measure fit to seen data rather than generalisation
- C. It always converts classification to regression
- D. It automatically creates more data

Which statement about scaling is most accurate?

- A. Every model always requires it
- B. No model can use it
- C. It helps many models but is not required for every model
- D. It replaces test evaluation

Part B — Practical Preprocessing Task (40 marks)

Use the customer churn dataset from the Week 4 Day 2 assignment or the dataset supplied by your trainer. Create Week4_Assessment_Preprocessing_YourName.ipynb.

1. Load the data with pandas. Display the first five rows, shape, data types, missing-value counts, duplicate count, and target distribution. (5 marks)
2. Identify the target and input features. Explain why the target must not be included in X. (5 marks)

3. Split into train and test sets before fitting any imputer, scaler, or encoder. Use a fixed random_state and explain the purpose. (5 marks)
4. Build a numeric pipeline with median imputation and scaling. (7 marks)
5. Build a categorical pipeline with missing-value imputation and OneHotEncoder(handle_unknown='ignore'). (7 marks)
6. Combine the steps using ColumnTransformer. Fit on training data only and transform both train and test data. (6 marks)
7. Print transformed shapes and explain what changed. Identify one possible leakage risk. (5 marks)

Part C — Debugging (15 marks)

1. Scaling before splitting: A student scales the full dataset before splitting into train and test.

Your task: Explain the risk and give the correct sequence. (3 marks)

Ans: The student should split the data first, then scale it using the training data. Scaling the entire dataset before splitting can affect the test results and make them less reliable.

2. Target leakage: A student includes churn in X and also uses it as y.

Your task: Explain why this invalidates the exercise and correct it. (3 marks)

Ans: The student should remove the churn column from X and use it only as y. Otherwise, the model already knows the answer it is supposed to predict, so the results will be misleading.

3. Unknown category: A test row contains a city not seen during training and the encoder raises an error.

Your task: Name the encoder setting that can help and explain it. (3 marks)

Ans: The student should use `OneHotEncoder(handle_unknown="ignore")` so the encoder can handle new cities that were not seen during training without raising an error.

4. Wrong imputation: A student fills a numeric column with the text 'unknown' and then scales it.

Your task: Explain the type mismatch and suggest a better strategy. (3 marks)

Ans: The student should fill missing numeric values with a number, such as the median, instead of the text 'unknown'. Otherwise, the scaler cannot process the column properly.

5. Misleading evaluation: A student uses training accuracy as proof the model works on new customers.

Your task: Explain why a held-out test set is needed. (3 marks)

Ans: The student should check the model's accuracy using test data, not just training data. This helps them understand how well the model predicts churn for new customers.

Part D — Explain in Your Own Words (10 marks)

8. Explain supervised vs. unsupervised learning, with one example. (2 marks)
   Ans: supervised learning uses labeled data to make predictions.
   Ex:predicting the customer will leave the shop.
   Unsupervised learning uses unlablled data.
   Ex: rouping customers based on their shopping behaviour.
9. Explain features vs. labels using one example. (2 marks)
   Ans: Features are the input data used to make predictions, while labels are the answers we want to predict. Example: customer age is a feature, and whether the customer leaves the company is the label.
10. Explain why preprocessing should be fitted on training data only. (2 marks)
    Ans: Preprocessing should be fitted only on training data to prevent data leakage and ensure reliable test results.

11. Explain fit vs. transform. (2 marks)
    Ans: fit learns information from the data, while `transform` applies the learned changes to the data.
12. Give one problem where a simple rule is preferable to ML. (2 marks)
    Ans: A simple rule is better for checking whether a person is eligible to vote because the decision can be made directly using the person's age.

Part E — Concept Viva (15 marks)

The trainer will ask you to explain your own notebook. Do not simply read your notes.

- Explain your workflow from loading data to transforming it.
- Point to one numeric feature and explain its imputation and scaling.
- Point to one categorical feature and explain how encoding changes it.
- Explain why the target is excluded from X.
- Explain how your code avoids fitting preprocessing on the test set.
- Describe one error you encountered or could encounter and how you would debug it.

Viva scoring: concepts (5), explaining code (5), debugging and reasoning (5).

Student Submission Checklist

- Quiz answers completed.
- Notebook runs top to bottom.
- Inspection outputs are visible.
- Train/test split happens before fitting transformations.
- Numeric and categorical pipelines are implemented.
- ColumnTransformer is used correctly.
- Debugging and own-words answers are included.
- I can explain my notebook without reading every line.

Trainer Answer Key — Remove Before Sharing Student Copy

MCQ answers: 1-A, 2-B, 3-C, 4-C, 5-B, 6-B, 7-B, 8-A, 9-B, 10-B, 11-A, 12-B, 13-B, 14-C, 15-A, 16-A, 17-B, 18-A, 19-B, 20-C.

Practical grading: reward accurate inspection, clear X/y separation, splitting before fitting transformations, median imputation and scaling for numeric columns, categorical imputation and one-hot encoding, correct ColumnTransformer use, and a clear explanation of leakage. Accept equivalent valid approaches.

The sample dataset is tiny: any model score is only a workflow demonstration, not reliable evidence of model quality.